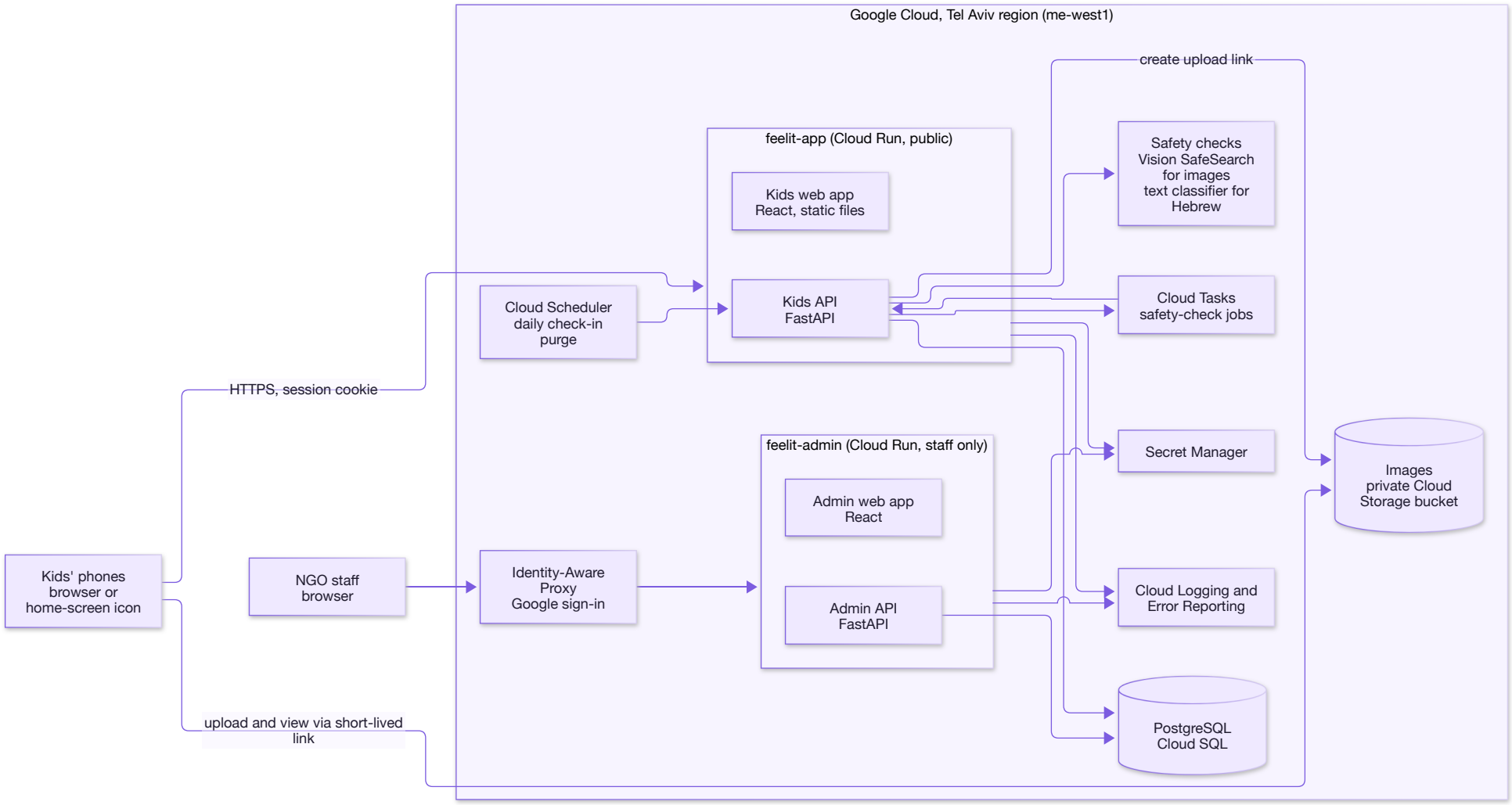
FeelIt · v1 system design

How the app is built and run for a pilot of about 10 schools and 2,000 kids, starting with one student developer and a mentor.

Status	Proposal for review
Date	November 2026
Builds on	`design/data-model.mmd` and `design/FeelIt-Data-Model.pdf`
Diagram sources	`design/system/*.mmd`
In short	Installable React web app · Python FastAPI · PostgreSQL · Google Cloud, Tel Aviv region

Architecture

One codebase · two Cloud Run services · data stays in Israel



Building blocks and why

Simple enough for a student, ready for a team

Kids web app

- **React + Vite + TypeScript**, Hebrew right-to-left, built for phone screens.
- **Installable** to the home screen (PWA): icon and full screen. No offline mode and no notifications.
- Served by the API container, on the **same domain** as the API, so no cross-origin setup is needed.

Admin web app

- React, for NGO staff: schools, teachers, codes, catalogs, help lines, settings, hidden items.
- Separate deployment behind **Google sign-in**; never reachable from the internet without it.

API

- **Python, FastAPI**, SQLAlchemy, Alembic for schema changes, Pydantic for request checks.
- **One service**, split into modules that follow the data model: access, check-in, giving, plant, polls, stories, reports, admin.
- About 2,000 kids is a small load, so there's no need for microservices, a cache or live connections.
- FastAPI generates the API reference (OpenAPI) from the code.

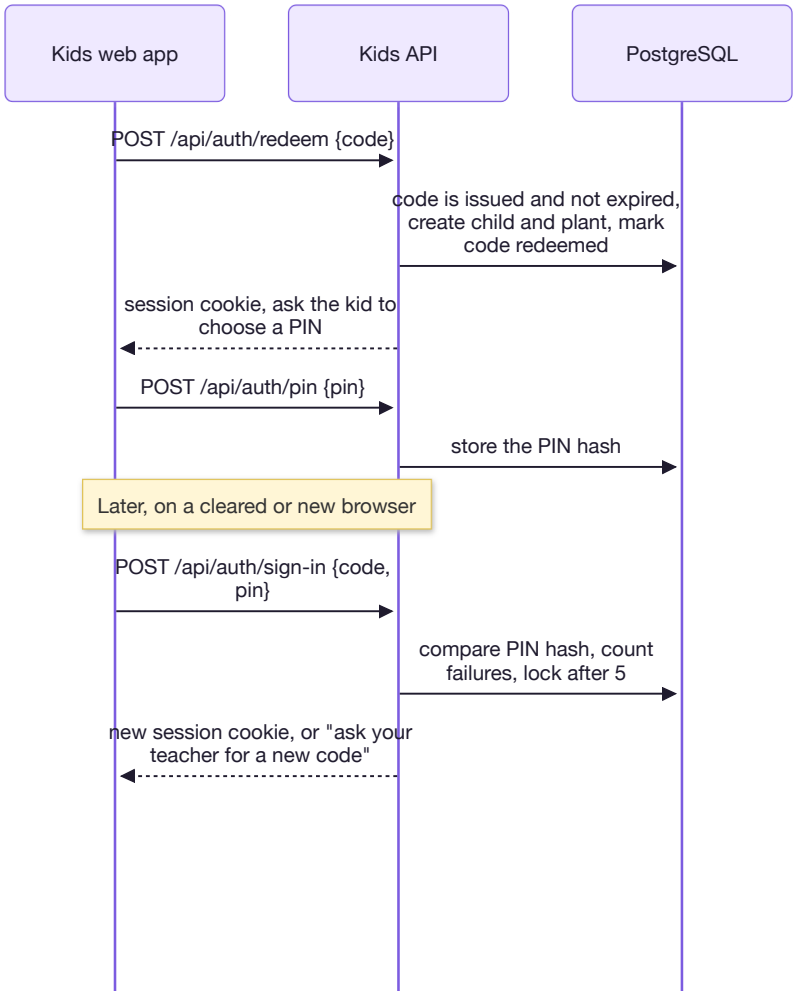
Google Cloud, Tel Aviv

- **Cloud Run:** runs the container, scales to zero, no servers to manage.
- **Cloud SQL (PostgreSQL)** with automatic backups.
- **Cloud Storage:** private image bucket; phones use short-lived upload and view links.
- **Cloud Tasks** for safety-check jobs; **Cloud Scheduler** for the daily check-in purge.
- **Vision SafeSearch** for images; a text classifier for Hebrew (to be tested).
- **Secret Manager**, and **Cloud Logging and Error Reporting** so logs stay in the project.

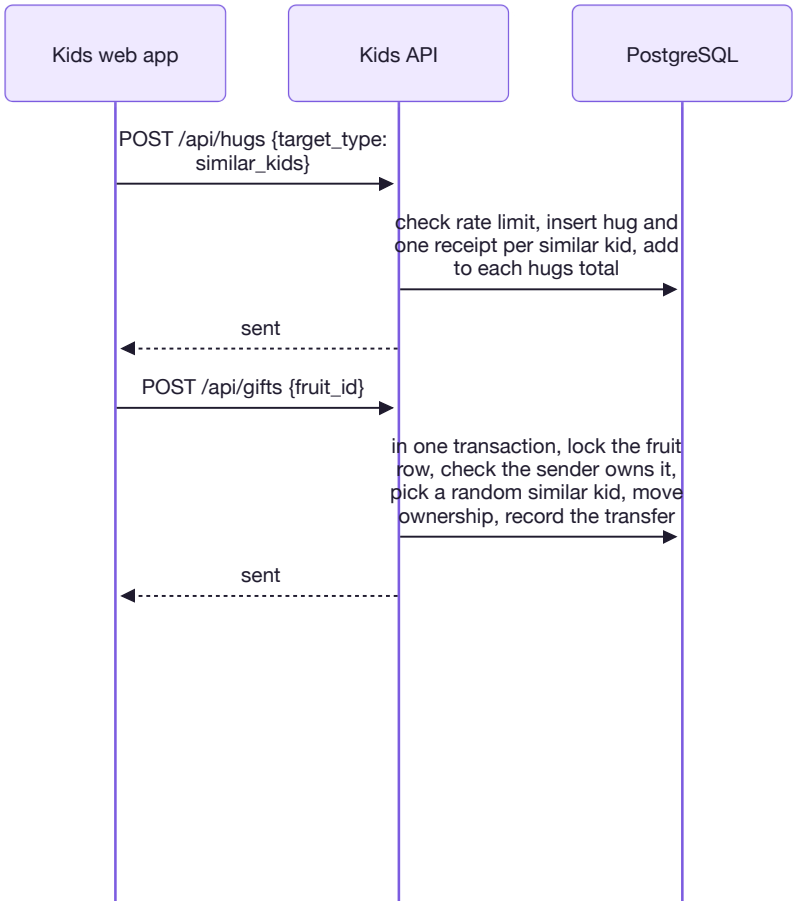
Signing in and giving

Teacher code · 6-digit PIN · gifts are one transaction

First sign-in, then signing in again



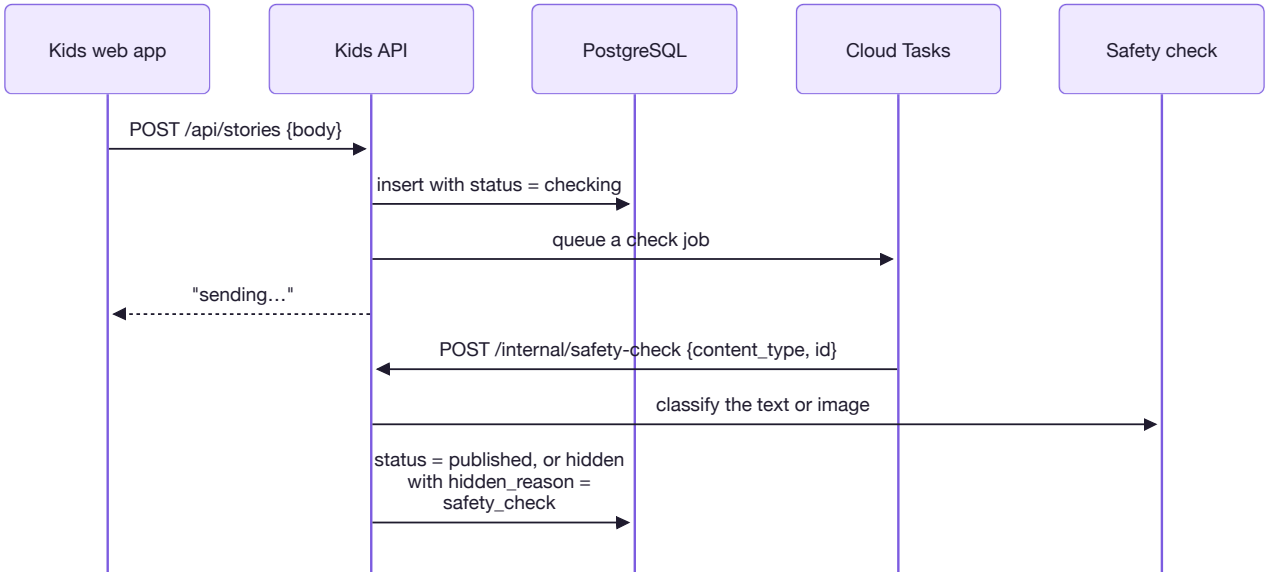
Group hug and fruit gift



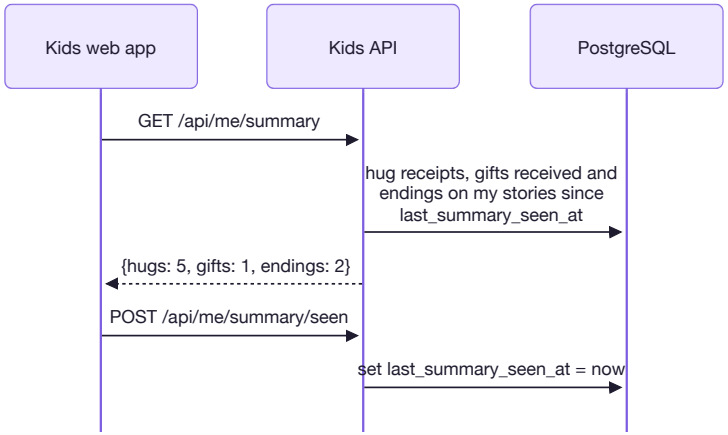
Posting content and coming back

Safety check before publishing · "while you were away"

Story, ending, poll or image



Opening the app



Kids API (draft)

Public service · session cookie · all paths under /api

Method	Path	What it does
Sign-in		
POST	/auth/redeem	Redeem a teacher's code; creates or relinks the child
POST	/auth/pin	Set the 6-digit PIN right after redeeming
POST	/auth/sign-in	Code plus PIN; locks after 5 wrong tries
POST	/auth/sign-out	End this session
Me and my plant		
GET	/me	First name, plant stage, water, hugs received, stickers
PUT	/me/stickers	Choose the stickers on my pot
GET	/me/fruits	My fruits, grown and received
GET	/me/stories	My stories, with their endings
GET	/me/polls	"My questions", with results
GET	/me/summary	While you were away: hugs, gifts, endings
POST	/me/summary/seen	Mark the summary as seen
Catalogs		
GET	/catalog	Sliders, stickers, fruit types and help lines, in one cacheable response

Method	Path	What it does
Check-in and giving		
GET	/check-ins/today	Today's check-in, if any
PUT	/check-ins/today	Save or edit today's slider values
GET	/similar	Similar-kids count (hidden below the minimum) and sample stickers
POST	/hugs	Hug the similar kids, a story, an ending or a poll
POST	/gifts	Give a fruit to a random similar kid, or a story's or ending's author
Polls and images		
GET	/polls	Poll feed (paged)
POST	/polls	New poll with 3–4 image options
POST	/polls/{id}/votes	Vote once; returns the results
GET	/images/library	Pre-approved images
POST	/images/uploads	Get an upload link and image ID
POST	/images/{id}/uploaded	Upload finished; start the safety check
Stories and reports		
GET	/stories	Story feed (paged)
POST	/stories	Share a story
GET	/stories/{id}	A story and its endings
POST	/stories/{id}/endings	Suggest a happy ending
POST	/reports	Report a poll, story, ending or image

Admin and internal API (draft)

Admin: staff only, behind Google sign-in · Internal: called by Google Cloud jobs

Method	Path (under /admin/api)	What it does
Schools, teachers and codes		
GET · POST	/schools	List and add schools
GET · POST	/schools/{id}/teachers	List and add a school's teachers
POST	/teachers/{id}/codes	Issue a batch of new codes, ready to print
GET	/teachers/{id}/codes	A teacher's codes and their status
POST	/codes/{code}/relink	New relink code for the child who redeemed this code
POST	/codes/{code}/revoke	Revoke a code and sign out its sessions
Content and configuration		
GET	/hidden	Hidden items with their reason and report count
CRUD	/sliders, /stickers, /fruit-types	Catalogs
CRUD	/library-images	Pre-approved poll images
CRUD	/help-lines	Help screen entries, with the date checked
GET · PUT	/settings	Similarity window, thresholds, limits, retention

Method	Path (under /internal)	Called by
POST	/safety-check	Cloud Tasks, once per new story, ending, poll or image
POST	/purge-check-ins	Cloud Scheduler, daily

Notes

- Internal endpoints accept only Google-signed requests from our own service account.
- Codes are tracked by code, not by child name, so staff never need to know who a child is. The teacher knows which paper code went to which kid.
- No admin endpoint shows check-ins or who wrote, voted, sent or reported anything.
- No endpoint exposes another kid's ID. Feeds return anonymous items only.

Security, privacy and operations

For a pilot with children's data

Sessions and sign-in

- The session is a random token in a **server-set cookie** that scripts can't read (`HttpOnly`, `Secure`, `SameSite=Lax`). Only its hash is stored.
- Server-set cookies are not cleared by Safari's 7-day limit on script-stored data.
- **PIN:** 6 digits, stored with a slow password hash (Argon2). Locks after 5 tries, plus a rate limit per IP.
- Codes are long and random, expire if unused, and are single-use for redeeming.

Privacy

- All data is stored in the **Tel Aviv region**, including backups and logs.
- Logs never include slider values, story text, PINs or names.
- No analytics in the pilot, and no third-party scripts in the app.
- Check-ins are deleted after the retention period; plant growth is kept separately.
- Staff access needs a Google account from the NGO's own list.

Building and running

- **GitHub Actions:** lint and test, run schema changes against a throwaway PostgreSQL, build the image, deploy to Cloud Run.
- Deploys sign in to Google Cloud **without a stored key** (Workload Identity Federation).
- **Production only**, plus local development with Docker Compose and made-up seed data.
- Cloud SQL daily backups with point-in-time recovery.
- Alerts on error rate and on failed safety-check jobs.

Risks and open items

For discussion

Risks

- **Hebrew text check:** it's unconfirmed whether Google's text moderation handles Hebrew well, or which models run in the Tel Aviv region. Test real examples; a prompted model such as Gemini on Vertex AI is the fallback.
- **No human review in v1:** anything the automatic check misses stays up until enough reports hide it.
- **Production only:** nowhere to test except with real kids' data. Mitigated by CI checks and local seed data.
- **Guessable PINs:** the lockout and rate limits are what keep code-plus-PIN sign-in safe.
- **Privacy law:** obligations for minors' emotional data in Israel, consent wording and database ownership need confirming with counsel or the NGO before the pilot.

Open items

- Domain name for the app and the admin tool.
- Which Google Workspace accounts get admin access.
- Limits on image size and type, and on story and ending length.
- How codes are printed and handed to teachers.
- Who looks at hidden items, and how often.
- Monthly budget ceiling for the cloud account.